

ROBERVAL, QC, Canada

WMO: 717280

Lat: 48.5230N

Lon: 72.2662W

Elev: 179

StdP: 99.20

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.8	(c) -27.2	(d) -35.7	(e) 0.1	(f) -29.5	(g) -33.2	(h) 0.2	(i) -26.8	(j) 13.1	(k) -12.5	(l) 11.6	(m) -12.2	(n) 3.2	(o) 250	(p) 0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	28.4	20.0	26.6	19.2	24.8	18.4	21.4	26.3	20.4	24.9	19.4	23.6	4.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.7	24.2	18.6	13.8	23.1	17.7	12.9	22.1	63.2	26.3	59.5	24.9	56.2	23.6	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.3	8.3	-33.5	32.0	3.1	2.1	-35.7	33.5	-37.5	34.7	-39.3	35.9	-41.5	37.4
			-33.7	23.3	3.0	1.1	-35.9	24.1	-37.6	24.7	-39.3	25.3	-41.5	26.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-15.3	-13.1	-6.5	1.7	10.2	16.0	18.6	17.4	13.0	6.0	-1.4	-9.8		
	DBStd	12.93	7.78	7.24	6.87	5.00	4.76	4.28	3.19	3.31	4.40	4.54	5.45	6.90		
	HDD10.0	3369	783	647	512	252	57	4	0	0	18	141	343	614		
	HDD18.3	5651	1041	880	769	499	257	96	36	57	169	382	592	873		
	CDD10.0	869	0	0	1	2	62	183	266	230	108	17	1	0		
	CDD18.3	109	0	0	0	0	4	24	43	29	8	1	0	0		
	CDH23.3	718	0	0	1	1	53	193	261	159	49	2	0	0		
	CDH26.7	152	0	0	0	0	12	48	54	29	8	0	0	0		
(14) Wind	WSAvg	3.8	3.9	4.0	4.2	4.0	3.6	3.5	3.3	3.4	3.6	3.8	4.0	3.8		
(15) Precipitation	PrecAvg	863	56	47	49	61	64	86	106	89	93	81	66	66		
	PrecMax	1100	147	116	96	116	126	139	248	250	217	158	108	121		
	PrecMin	688	12	18	18	15	13	37	42	36	38	15	12	17		
	PrecStd	104	27	22	22	22	25	29	47	43	39	32	23	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	6.4	11.6	18.4	28.3	30.7	30.6	29.5	27.6	21.9	14.4	5.9		
		MCWB	6.0	3.9	6.2	10.3	17.0	20.6	21.1	21.1	20.5	15.2	11.3	4.8		
	2%	DB	2.4	2.2	7.0	14.7	24.3	27.9	28.1	27.1	24.5	18.3	10.7	3.2		
		MCWB	1.5	0.6	3.2	8.2	15.1	19.0	20.4	20.0	19.2	13.8	8.4	1.9		
	5%	DB	-0.4	0.2	4.6	11.4	21.2	25.5	26.2	25.0	22.0	15.4	8.1	1.2		
		MCWB	-1.4	-1.2	1.8	6.2	13.7	18.2	19.4	19.0	17.8	11.9	5.8	0.1		
	10%	DB	-3.7	-1.9	2.7	8.7	18.7	23.2	24.4	23.2	20.1	12.9	5.6	-0.2		
		MCWB	-4.7	-3.2	0.5	4.5	12.4	16.8	18.7	18.1	16.3	10.1	3.9	-1.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	4.5	8.0	12.4	18.8	22.3	23.0	22.2	21.5	16.5	11.7	4.6		
		MCDB	7.0	6.1	9.2	16.8	25.5	28.6	27.8	27.7	25.9	20.2	13.9	5.4		
	2%	WB	1.6	0.8	3.7	8.6	16.7	20.5	21.5	20.8	19.9	14.1	8.8	2.0		
		MCDB	2.4	2.1	6.3	13.1	21.8	25.6	26.1	25.7	23.6	17.9	10.5	3.0		
	5%	WB	-1.4	-1.1	2.1	6.7	14.6	19.2	20.4	19.8	18.3	12.2	6.1	0.3		
		MCDB	-0.4	0.3	4.2	10.8	19.9	24.0	25.0	23.7	21.8	14.8	7.7	1.1		
	10%	WB	-4.6	-3.2	0.6	5.0	12.7	17.8	19.4	18.8	16.7	10.3	4.0	-1.0		
		MCDB	-3.7	-1.9	2.6	8.3	18.0	22.2	23.6	22.3	19.6	12.4	5.4	-0.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.6	9.5	9.1	11.0	10.3	9.3	9.2	9.2	7.6	6.7	7.7		
		MCDBR	11.4	9.4	10.4	13.4	15.1	12.5	11.2	11.2	11.3	11.2	9.7	8.0		
	5% WB	MCWBR	11.0	8.5	8.1	8.7	8.3	7.1	6.1	6.4	7.2	7.7	8.2	7.7		
		MCWBR	11.4	9.3	9.4	12.1	13.2	11.0	10.0	9.6	10.4	10.1	9.4	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.280	0.303	0.371	0.352	0.383	0.395	0.374	0.362	0.329	0.314	0.259		
	taud		2.235	2.225	2.238	2.257	2.431	2.367	2.352	2.413	2.432	2.505	2.372	2.318		
	Ebn at Noon		793	874	911	876	909	879	861	866	841	817	735	765		
	Edh at Noon		75	97	114	125	109	118	118	106	95	75	68	61		
(44) All-Sky Solar Radiation	RadAvg		1.32	2.31	3.62	4.57	5.20	5.59	5.16	4.42	3.38	1.95	1.05	0.86		
	RadStd		0.17	0.30	0.32	0.39	0.48	0.40	0.52	0.28	0.27	0.19	0.12	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page